PBR VISVODAYA INSTITUTE OF TECHNOLOGY & SCIENCE

CLOUD COMPUTING & VIRTUALIZATION

Course Code: **23CS403** • Semester: **3-2** • Department of Computer Science & Engineering

Course Scope & Objectives: Cloud deployment models, virtualization architectures, hypervisors, cloud storage systems, microservices, containerization with Docker & Kubernetes, and serverless computing.

Unit 1: Cloud Computing Principles & Architecture

Key Syllabus Topics: NIST Cloud Definition, Essential Characteristics (On-demand Self-Service, Broad Network Access, Resource Pooling, Rapid Elasticity, Measured Service), Service Models (IaaS, PaaS, SaaS), Deployment Models (Public, Private, Hybrid, Community Cloud).

Theoretical Framework & Concepts:

Cloud computing shifts capital expenditure (CapEx) to operational expenditure (OpEx) through shared, elastic multi-tenant infrastructure.

Implementation / Key Formula Reference:

```
// Service Layering: SaaS (Applications) -> PaaS (Runtimes) -> IaaS (Raw Compute/Storage)
```

Frequently Asked University Exam Questions:

• Q1: Explain the NIST 5 Essential Characteristics of Cloud Computing.

Solution: A: On-demand self-service, Broad network access, Resource pooling (multi-tenancy), Rapid elasticity (auto-scaling), and Measured service (pay-per-use billing).

Unit 2: Virtualization Technologies & Hypervisors

Key Syllabus Topics: Virtualization Fundamentals, Type-1 (Bare-Metal) vs Type-2 (Hosted) Hypervisors, Full Virtualization (Binary Translation), Paravirtualization (Hypercalls), Hardware-Assisted Virtualization (Intel VT-x), Containerization vs Virtual Machines.

Theoretical Framework & Concepts:

Hypervisors virtualize physical hardware. Containers share the host kernel while isolating namespaces and cgroups, achieving sub-second startup and near-native execution speed.

Implementation / Key Formula Reference:

```
// VM: Guest OS on Hypervisor | Container: App & Dependencies on Shared OS Kernel
```

Frequently Asked University Exam Questions:

• Q1: Compare Virtual Machines and Docker Containers.

Solution: A: VMs bundle a complete Guest OS (heavyweight, gigabytes in size, slow boot); Containers share the host OS kernel (lightweight, megabytes in size, instant boot).

Unit 3: Cloud Storage & Distributed File Systems

Key Syllabus Topics: Cloud Storage Architectures (Block Storage, File Storage, Object Storage - S3), CAP Theorem (Consistency, Availability, Partition Tolerance), Distributed Hash Tables (DHT), Consistency Models (Eventual Consistency), Multi-Region Replication.

Theoretical Framework & Concepts:

Object storage stores unstructured data as objects with unique IDs and rich metadata accessed via REST APIs. The CAP theorem states a distributed system can guarantee at most two of C, A, and P.

Implementation / Key Formula Reference:

```
// CAP Theorem: In presence of network partition (P), choose between Consistency (C) or Availability (A)
```

Frequently Asked University Exam Questions:

• Q1: Explain CAP theorem in distributed cloud systems.

Solution: A: Network partitions are inevitable in real networks. Systems must choose between Consistency (all nodes return latest data) and Availability (every request receives non-error response).

Unit 4: Cloud Security, IAM, and Compliance

Key Syllabus Topics: Cloud Shared Responsibility Model, Identity and Access Management (IAM), Role-Based Access Control, Cloud Encryption at Rest & In-Transit, Key Management Services (KMS), Security Groups & Network ACLs, Disaster Recovery Strategies.

Theoretical Framework & Concepts:

In the Shared Responsibility Model, the cloud provider secures the cloud infrastructure (hardware, facilities, host virtualization), while the customer secures data, IAM policies, and OS configurations.

Implementation / Key Formula Reference:

```
{
  "Effect": "Allow",
  "Action": "s3:GetObject",
  "Resource": "arn:aws:s3:::pbrvits-notes/*"
}
```

Frequently Asked University Exam Questions:

• Q1: Explain the Cloud Shared Responsibility Model for IaaS vs SaaS.

Solution: A: In IaaS, provider manages hardware/network, customer manages OS, middleware, and apps. In SaaS, provider manages everything including software; customer only manages data and access credentials.

Unit 5: Cloud Native, Microservices, and Serverless

Key Syllabus Topics: Microservices Architecture vs Monoliths, Docker Containerization, Kubernetes (Pods, Services, Deployments, Auto-scaling), Serverless Computing (Function-as-a-Service - FaaS / AWS Lambda), Event-Driven Architectures, Edge Computing.

Theoretical Framework & Concepts:

Serverless FaaS automatically scales compute functions from zero to thousands of instances based on incoming triggers, eliminating server management overhead and idle costs.

Implementation / Key Formula Reference:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: campus-backend
spec:
  replicas: 3
```

Frequently Asked University Exam Questions:

• Q1: What is Serverless Computing (FaaS)? State its key benefits.

Solution: A: Event-driven execution model where cloud provider dynamically manages compute allocations. Benefits: Zero idle costs, automatic horizontal scaling, no server maintenance.